Shah Md. Tasrif Rahman

Assignment 3

SPSS For Research

Instructor: Md. Ahsanul Islam

Deadline: 29 August 2026

Submission: Save your assignment as a word document named Assignment 3_YOURNAME.docx, write your name (as per certificate) at the top, and submit the word file.

Problem Statement:

The dataset assignment3.sav contains records for 400 mothers who recently gave birth, collected in a maternal and child health survey. For each mother the file records place of residence, household wealth, age, the number of antenatal care (ANC) visits made during pregnancy, and whether the baby was born with low birth weight (below 2500 g). Use this one file for every question below.

Variable	Label	Values / Units	Measure
mother_id	Mother identification number	Integer ID	Nominal
low_birth_weight	Baby born with low birth weight	0 = Normal (≥2500 g); 1 = Low (<2500 g)	Nominal
mother_age	Mother's age	Years	Scale
anc_visits	Number of antenatal care (ANC) visits during pregnancy	Count	Scale
wealth_index	Household wealth index	1 = Poor; 2 = Middle; 3 = Rich	Ordinal
residence	Place of residence	0 = Rural; 1 = Urban	Nominal

Question 1

Produce descriptive statistics for the sample and complete **Table 1**. For the categorical variables report the frequency and percentage of each category; for the two continuous variables report the mean (standard deviation), median, minimum and maximum.

Table 1. Descriptive characteristics of the sample (N = 400).

Variable	Level / statistic	Frequency, n (%)	Mean (SD)	Median	Minimum	Maximum
Low birth weight	Normal (≥2500 g)	76.25%	—	—	—	—
	Low (<2500 g)	23.75%	—	—	—	—
Residence	Rural	61.00%	—	—	—	—
	Urban	39.00%	—	—	—	—
Wealth index	Poor	42.25%	—	—	—	—
	Middle	40.50%	—	—	—	—
	Rich	17.25%	—	—	—	—
Mother's age (years)		—	25.97 (4.70)	26	16	39
ANC visits		—	4.64 (2.07)	5	0	11

Cells marked — are not applicable to that variable and should be left blank.

Question 2

Test whether the mean number of ANC visits differs between mothers living in rural and urban areas. State the null and alternative hypotheses, then run Independent-Samples T Test with anc_visits as the Test Variable and residence as the Grouping Variable (Define Groups: 0 and 1). Use Levene's test to decide which row to read, complete Table 2, and write a one-sentence conclusion in APA style.

Table 2a. Group statistics for ANC visits by residence.

Residence group	n	Mean	Std. Deviation	Std. Error Mean
Rural	244	3.9713	1.88212	0.12049
Urban	156	5.6923	1.91637	0.15343

Table 2b. Independent-samples t-test result.

Independent-samples t-test statistic	**Value**
Levene's Test for Equality of Variances - F	.131
Levene's Test P-value	.717
t	-8.857
df	398
P-value	< .001
Mean difference (Urban - Rural)	-1.721
Std. error difference	0.19431
95% Confidence Interval	[-2.10301, -1.33899]

Here, for the independent-samples t-test,

H_0: The mean number of ANC visits does not differ between mothers living in rural and urban areas. ($\mu_1 = \mu_2$)

H_1: The mean number of ANC visits differs between mothers living in rural and urban areas. ($\mu_1 \neq \mu_2$).

<u>Conclusion</u>: An independent-samples t-test was conducted to compare the mean number of ANC visits between mothers living in rural and urban areas. There was a significant difference in the mean number of ANC visits for mothers living in rural areas (*M* = 3.9713, *SD* = 1.88212) and mothers living in urban areas (*M* = 5.6923, *SD* = 1.91637), *t*(398) = -8.857, *p* = < .001, *d* = 1.89553.

Question 3

Fit a binary logistic regression predicting low birth weight from mother's age, number of ANC visits, wealth index and residence.

Complete Table 3 from the SPSS's output and also explain the relationships between the independent variables and low birth weight.

Table 3. Binary logistic regression predicting low birth weight (1 = Low).

Predictor	Estimate (B)	SE	Odds ratio Exp(B)	95% CI for OR	p-value
Mother's age	-.064	.028	.938	[.888, .990]	.021
ANC visits	-.244	.070	.783	[.682, .899]	< .001
Wealth index: Poor	— reference —				
Wealth index: Middle	-.727	.274	.483	[.282, .827]	.008
Wealth index: Rich	-.923	.434	.397	[.170, .929]	.033
Residence: Rural	— reference —				
Residence: Urban	-.656	.311	.519	[.282, .955]	.035
Constant	2.126	.787	8.378		.007

Reference categories are marked. Also report the overall model fit below the table:

Omnibus test of model coefficients: $\chi^2(5) = 50.129$, p = < .001. −2 Log-likelihood = 388.416. Nagelkerke $R^2 = 0.177$.

Hosmer–Lemeshow test: $\chi^2(8) = 5.653$, p = .686. Overall percentage correctly classified = 17.7%.

<u>Conclusion</u>: A binary logistic regression was conducted to examine whether mother's age, number of antenatal care (ANC) visits, household wealth index, and place of residence predicted whether the baby was born with low birth weight (1 = low, 0 = normal). Poor wealth status and rural residence were used as the reference categories for the categorical predictors.

The overall model was statistically significant, $\chi^2(5) = 50.129$, $p < .001$, with a Nagelkerke R^2 of .177, indicating that the model provided a pseudo-R^2 of approximately 17.7%. The Hosmer–Lemeshow test was not significant, $\chi^2(8) = 5.653$, $p = .686$, indicating no evidence of lack of fit. Maternal age was significantly associated with low birth weight, $B = -0.064$, $SE = 0.028$, $OR = 0.938$, 95% CI [0.888, 0.990], $p = .021$, such that each additional year of maternal age was associated with a 6.2% decrease in the odds of low birth weight. ANC visits were also significantly associated with low birth weight, $B = -0.244$, $SE = 0.070$, $OR = 0.783$, 95% CI [0.682, 0.899], $p < .001$; each additional ANC visit was associated with a 21.7% decrease in the odds of low birth weight. Compared with mothers from poor households, those from middle-wealth households had 51.7% lower odds of low birth weight, $OR = 0.483$, 95% CI [0.282,

0.827], *p* = .008, while those from rich households had 60.3% lower odds, *OR* = 0.397, 95% CI [0.170, 0.929], *p* = .033. Finally, urban residence was associated with lower odds of low birth weight compared with rural residence, *OR* = 0.519, 95% CI [0.282, 0.955], *p* = .035. Overall, greater maternal age, more ANC visits, higher household wealth, and urban residence were associated with lower odds of low birth weight.

Appendix:

Appendix-A:

```
* Encoding: UTF-8.

* /Step-1: Data loading/

GET
  FILE='E:\Statistical Softwares\SPSS\CUSPC-SPSS\Assignments\Assignment-
3\assignment3.sav'.

Warning # 67.  Command name: GET FILE
The document is already in use by another user or process.  If you make
changes to the document they may overwrite changes made by others or your
changes may be overwritten by others.
File opened E:\Statistical Softwares\SPSS\CUSPC-SPSS\Assignments\Assignment-
3\assignment3.sav
DATASET NAME DataSet1 WINDOW=FRONT.
```

Dataset Name

Warnings
The active dataset will replace the existing dataset named DataSet1.

```
DATASET ACTIVATE DataSet1.

* /Step-2: Data Checking/

FREQUENCIES VARIABLES=mother_id low_birth_weight mother_age anc_visits wealth_index
residence
/FORMAT=NOTABLE
/STATISTICS=RANGE MINIMUM MAXIMUM
/ORDER=ANALYSIS.
```

Frequencies

Statistics

		Mother identification number	Baby born with low birth weight (<2500 g)	Mother's age (years)	Number of antenatal care (ANC) visits during pregnancy	Household wealth index	Place of residence
N	Valid	400	400	400	400	400	400
	Missing	0	0	0	0	0	0
Range		399.00	1.00	23.00	11.00	2.00	1.00
Minimum		1001.00	.00	16.00	.00	1.00	.00
Maximum		1400.00	1.00	39.00	11.00	3.00	1.00

```
* No missing or impossible values exist, we can continue.

* Question 1:

* Custom Tables.
CTABLES
  /VLABELS VARIABLES=low_birth_weight residence wealth_index mother_age anc_visits
DISPLAY=LABEL
  /TABLE low_birth_weight [TABLEPCT.COUNT 'Frequency, n (%)' PCT40.2] + residence
    [TABLEPCT.COUNT 'Frequency, n (%)' PCT40.2] + wealth_index [TABLEPCT.COUNT 'Frequency,
n (%)'
    PCT40.2] + mother_age [MEAN, STDDEV, MEDIAN, MINIMUM, MAXIMUM] + anc_visits [MEAN,
STDDEV, MEDIAN,
    MINIMUM, MAXIMUM]
  /CATEGORIES VARIABLES=low_birth_weight residence wealth_index ORDER=A KEY=VALUE
EMPTY=INCLUDE
  /CRITERIA CILEVEL=95.
```

Custom Tables

		Frequency, n (%)	Mean	Standard Deviation	Median	Minimum	Maximum
Baby born with low birth weight (<2500 g)	Normal (>=2500 g)	76.25%					
	Low (<2500 g)	23.75%					
Place of residence	Rural	61.00%					
	Urban	39.00%					
Household wealth index	Poor	42.25%					
	Middle	40.50%					
	Rich	17.25%					
Mother's age (years)			25.97	4.70	26.00	16.00	39.00
Number of antenatal care (ANC) visits during pregnancy			4.64	2.07	5.00	.00	11.00

```
* Question 2:

* H_0: The mean number of ANC visits doesn't differ between mothers living in rural and
urban areas. (μ_1 = μ_2).
* H_1: The mean number of ANC visits differs between mothers living in rural and urban
areas. (μ_1 ≠ μ_2).

T-TEST GROUPS=residence(0 1)
  /MISSING=ANALYSIS
  /VARIABLES=anc_visits
  /ES DISPLAY(TRUE)
  /CRITERIA=CI(.95).
```

T-Test

Group Statistics

	Place of residence	N	Mean	Std. Deviation	Std. Error Mean
Number of antenatal care (ANC) visits during pregnancy	Rural	244	3.9713	1.88212	.12049
	Urban	156	5.6923	1.91637	.15343

Independent Samples Test

		Levene's Test for Equality of Variances		t-test for Equality of Means						
									95% Confidence Interval of the Difference	
		F	Sig.	t	df	Sig. (2-tailed)	Mean Difference	Std. Error Difference	Lower	Upper
Number of antenatal care (ANC) visits during pregnancy	Equal variances assumed	.131	.717	-8.857	398	.000	-1.72100	.19431	-2.10301	-1.33899
	Equal variances not assumed			-8.822	326.034	.000	-1.72100	.19509	-2.10479	-1.33721

Independent Samples Effect Sizes

		Standardizer[a]	Point Estimate	95% Confidence Interval	
				Lower	Upper
Number of antenatal care (ANC) visits during pregnancy	Cohen's d	1.89553	-.908	-1.118	-.697
	Hedges' correction	1.89911	-.906	-1.116	-.696
	Glass's delta	1.91637	-.898	-1.121	-.672

a. The denominator used in estimating the effect sizes.

Cohen's d uses the pooled standard deviation.

Hedges' correction uses the pooled standard deviation, plus a correction factor.

Glass's delta uses the sample standard deviation of the control group.

```
* Question 3:

LOGISTIC REGRESSION VARIABLES low_birth_weight
  /METHOD=ENTER mother_age anc_visits wealth_index residence
  /CONTRAST (wealth_index)=Indicator(1)
  /CONTRAST (residence)=Indicator(1)
  /CLASSPLOT
  /PRINT=GOODFIT CI(95)
  /CRITERIA=PIN(0.05) POUT(0.10) ITERATE(20) CUT(0.5).
```

Logistic Regression

Case Processing Summary

Unweighted Cases[a]		N	Percent
Selected Cases	Included in Analysis	400	100.0
	Missing Cases	0	.0
	Total	400	100.0
Unselected Cases		0	.0
Total		400	100.0

a. If weight is in effect, see classification table for the total number of cases.

Dependent Variable Encoding

Original Value	Internal Value
Normal (>=2500 g)	0
Low (<2500 g)	1

Categorical Variables Codings

			Parameter coding	
		Frequency	(1)	(2)
Household wealth index	Poor	169	.000	.000
	Middle	162	1.000	.000
	Rich	69	.000	1.000
Place of residence	Rural	244	.000	
	Urban	156	1.000	

Block 0: Beginning Block

Classification Table[a,b]

Observed			Predicted: Baby born with low birth weight (<2500 g): Normal (>=2500 g)	Predicted: Baby born with low birth weight (<2500 g): Low (<2500 g)	Percentage Correct
Step 0	Baby born with low birth weight (<2500 g)	Normal (>=2500 g)	305	0	100.0
		Low (<2500 g)	95	0	.0
	Overall Percentage				76.3

a. Constant is included in the model.

b. The cut value is .500

Variables in the Equation

		B	S.E.	Wald	df	Sig.	Exp(B)
Step 0	Constant	-1.166	.117	98.556	1	.000	.311

Variables not in the Equation

			Score	df	Sig.
Step 0	Variables	Mother's age (years)	4.723	1	.030
		Number of antenatal care (ANC) visits during pregnancy	26.734	1	.000
		Household wealth index	19.116	2	.000
		Household wealth index(1)	5.143	1	.023
		Household wealth index(2)	6.804	1	.009
		Place of residence(1)	18.906	1	.000
	Overall Statistics		46.002	5	.000

Block 1: Method = Enter

Omnibus Tests of Model Coefficients

		Chi-square	df	Sig.
Step 1	Step	50.129	5	.000
	Block	50.129	5	.000
	Model	50.129	5	.000

Model Summary

Step	-2 Log likelihood	Cox & Snell R Square	Nagelkerke R Square
1	388.416[a]	.118	.177

a. Estimation terminated at iteration number 5 because parameter estimates changed by less than .001.

Hosmer and Lemeshow Test

Step	Chi-square	df	Sig.
1	5.653	8	.686

Contingency Table for Hosmer and Lemeshow Test

		Baby born with low birth weight (<2500 g) = Normal (>=2500 g)		Baby born with low birth weight (<2500 g) = Low (<2500 g)		
		Observed	Expected	Observed	Expected	Total
Step 1	1	41	39.056	0	1.944	41
	2	37	36.657	3	3.343	40
	3	34	36.251	7	4.749	41
	4	34	34.075	6	5.925	40
	5	34	32.509	6	7.491	40
	6	31	31.701	10	9.299	41
	7	25	28.536	15	11.464	40
	8	26	25.839	14	14.161	40
	9	25	23.228	15	16.772	40
	10	18	17.148	19	19.852	37

Classification Table[a]

Observed			Predicted: Baby born with low birth weight (<2500 g) — Normal (>=2500 g)	Predicted: Baby born with low birth weight (<2500 g) — Low (<2500 g)	Percentage Correct
Step 1	Baby born with low birth weight (<2500 g)	Normal (>=2500 g)	297	8	97.4
		Low (<2500 g)	81	14	14.7
	Overall Percentage				77.8

a. The cut value is .500

Variables in the Equation

		B	S.E.	Wald	df	Sig.	Exp(B)	95% C.I.for EXP(B) Lower	95% C.I.for EXP(B) Upper
Step 1[a]	Mother's age (years)	-.064	.028	5.326	1	.021	.938	.888	.990
	Number of antenatal care (ANC) visits during pregnancy	-.244	.070	12.012	1	.001	.783	.682	.899
	Household wealth index			9.272	2	.010			
	Household wealth index(1)	-.727	.274	7.044	1	.008	.483	.282	.827
	Household wealth index(2)	-.923	.434	4.532	1	.033	.397	.170	.929
	Place of residence(1)	-.656	.311	4.450	1	.035	.519	.282	.955
	Constant	2.126	.787	7.291	1	.007	8.378		

a. Variable(s) entered on step 1: Mother's age (years), Number of antenatal care (ANC) visits during pregnancy, Household wealth index, Place of residence.

```
              Step number: 1

              Observed Groups and Predicted Probabilities

      20 +
+
         I
I
         I        L
I
F        I        N    L
I
R     15 +        N   LL
+
E        I        N   LL    L
I
Q        I       NNL  NN L  N  L L
I
U        I      NNNN  NN N  N  L L
I
E     10 +     NNNNNLLNN N  N  N N             L
+
N        I     NNNNNNLNNNN  N LNLN             L
I
C        I   NNNNNNNNNNNNNNLLNNLNNN   LLL     LLN   L       L
I
Y        I   NNNNNNNNNNNNNNLNNNNNNNNL  LNN    LLLNLL L       L
I
       5 +  NNNNNNNNNNNNNNNNNNNNNNNL LLNNLL LNLLNLL N L  N  N
+
         I  NNNNNNNNNNNNNNNNNNNNNNNNNLLNNNLNLLNNNNLLLN LL N  N
I
         I  NNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNLNLNNL N LLLL
I
         I NNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNL N NNLNL  LLNN NL
I
Predicted ---------+---------+---------+---------+---------+---------+---------+---------
+---------+----------
  Prob:   0       .1        .2        .3        .4        .5        .6        .7        .8
.9        1
  Group:
NNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLL
LLLLLLLLL

          Predicted Probability is of Membership for Low (<2500 g)
          The Cut Value is .50
          Symbols: N - Normal (>=2500 g)
                   L - Low (<2500 g)
          Each Symbol Represents 1.25 Cases.
```

<u>Appendix-B</u>:

```
* Encoding: UTF-8.


* /Step-1: Data loading/


GET

  FILE='E:\Statistical Softwares\SPSS\CUSPC-SPSS\Assignments\Assignment-

3\assignment3.sav'.

DATASET NAME DataSet1 WINDOW=FRONT.
```

```
DATASET ACTIVATE DataSet1.

* /Step-2: Data Checking/

FREQUENCIES VARIABLES=mother_id low_birth_weight mother_age anc_visits wealth_index
residence
/FORMAT=NOTABLE
/STATISTICS=RANGE MINIMUM MAXIMUM
/ORDER=ANALYSIS.

* No missing or impossible values exist, we can continue.

* Question 1:

* Custom Tables.
CTABLES
  /VLABELS VARIABLES=low_birth_weight residence wealth_index mother_age anc_visits
DISPLAY=LABEL
  /TABLE low_birth_weight [TABLEPCT.COUNT 'Frequency, n (%)' PCT40.2] + residence
    [TABLEPCT.COUNT 'Frequency, n (%)' PCT40.2] + wealth_index [TABLEPCT.COUNT 'Frequency,
n (%)'
    PCT40.2] + mother_age [MEAN, STDDEV, MEDIAN, MINIMUM, MAXIMUM] + anc_visits [MEAN,
STDDEV, MEDIAN,
    MINIMUM, MAXIMUM]
  /CATEGORIES VARIABLES=low_birth_weight residence wealth_index ORDER=A KEY=VALUE
EMPTY=INCLUDE
  /CRITERIA CILEVEL=95.

* Question 2:

* H_0: The mean number of ANC visits doesn't differ between mothers living in rural and
urban areas. (μ_1 = μ_2).
* H_1: The mean number of ANC visits differs between mothers living in rural and urban
areas. (μ_1 ≠ μ_2).
```

```
T-TEST GROUPS=residence(0 1)
  /MISSING=ANALYSIS
  /VARIABLES=anc_visits
  /ES DISPLAY(TRUE)
  /CRITERIA=CI(.95).

* Question 3:

LOGISTIC REGRESSION VARIABLES low_birth_weight
  /METHOD=ENTER mother_age anc_visits wealth_index residence
  /CONTRAST (wealth_index)=Indicator(1)
  /CONTRAST (residence)=Indicator(1)
  /CLASSPLOT
  /PRINT=GOODFIT CI(95)
  /CRITERIA=PIN(0.05) POUT(0.10) ITERATE(20) CUT(0.5).
```